CUSTOMER MILESTONE BRIEF

MS #14 - Verified Semantic Memory

Purpose, customer value, delivered capability, evidence, and claim boundaries

Status	Complete
Release	0.14.0
Date	2026-08-14
Audience	Customers, partners, executives, architects, delivery teams, and assurance reviewers

Executive summary

MS #14 - Verified Semantic Memory converts a specific part of the LIGHTYEAR modernization approach into a governed, reviewable capability. Allows the factory to reuse evidence-bound experience without allowing stale, failed, or private verifier information to contaminate future work.

Why this matters to a customer

Allows the factory to reuse evidence-bound experience without allowing stale, failed, or private verifier information to contaminate future work.

The practical result is a narrower, evidence-backed decision instead of a broad modernization claim. Commercial, architecture, delivery, security, and assurance teams can review the same capability, proof, and limits.

What the milestone delivers

- Added controller-owned, content-addressed semantic experiences that bind verified plans, patches, outcomes, graph nodes, source-evidence capsules, paths, gates, and run identities.
- Added positive repair, correct-unchanged, and non-executable negative memory classes; controller failures and unapproved evidence classes are quarantined instead of becoming reusable knowledge.
- Added a hard contamination boundary that excludes sealed-holdout runs and verifier-private artifacts from implementer memory, with adversarial validation and read-only projections.
- Added graph-, evidence-, path-, and vocabulary-aware retrieval with independent byte and result limits; graph or evidence-pack changes immediately stale prior experiences.
- Added progressive planner and builder memory context, while preserving fresh source evidence and deterministic gates as higher-authority inputs.
- Added versioned experience, snapshot, retrieval, and policy schemas plus cross-platform memory launchers, CLI ingestion/query/validation, and a Verified Experience Memory dashboard.

- Bound the verified-memory snapshot into the hash-chained audit ledger and current release dossier; full verification now rejects stale, tampered, or privacy-contaminated memory.
- Added tamper, idempotency, invalidation, negative-replay, sealed-contamination, context-budget, API, UI, and full-suite regression coverage.

How it differs from earlier work

MS #14 builds on MS #13 (Sealed Quality Gates). The earlier milestone established its own bounded capability; this milestone adds verified semantic memory while preserving the earlier evidence and its limits.

Earlier evidence is not automatically promoted into this milestone. A parser, simulator, local comparison, or generated artifact is not live platform equivalence or production readiness.

What a customer can do with it

A customer can use verified semantic memory as a bounded decision and delivery capability, attached to approved sources, owners, policies, target architecture, acceptance criteria, and authorized evidence.

Unresolved semantic loss, policy decisions, unsupported behavior, and live-evidence requirements remain visible.

Evidence and acceptance posture

The repository capability is complete because its committed implementation and release record are present. Customer qualification remains claim-specific and evidence-class-specific.

Review exact source and artifact identities, distinguish development from authorized live evidence, inspect policy and loss classifications, confirm passed and blocked gates, and verify that later changes have not invalidated the evidence.

Boundaries and claims not made

- This milestone establishes only the bounded capabilities and evidence listed here; later capabilities are not retroactive.
- Production readiness and platform equivalence require the explicit evidence and policy gates applicable to the customer environment.

Source of record

- CHANGELOG.md release section(s): 0.14.0
- README.md product and operating guidance
- LIGHTYEAR-ROADMAP.md governing sequence and claim classifications
- docs/milestones/catalog.json metadata and docs/milestones/manifest.json artifact integrity